

MB 522: Machine Learning for Business

Course Code	MB 522
Course Name	Machine Learning for Business
L-T-P-C	2-0-0-2
Prerequisites	None
Intended for	MBA
Mutual Exclusion	None
Approval	52nd BoA (Nov 2023)

Course Contents

- **Introduction to Machine Learning:** Understanding what is ML, its purpose, how and when to use it in Business, ML and Statistics-similarities and contrasts, Data driven decision making, ML methods. (3 Lectures)
- **Machine Learning and Data Analytics:** Data Models- Linear vs. Non-linear, complicated vs. Complex models for real world problem solving; ML system requirements, Iterative ML development Process, Framing ML Problems; feature understanding and selection. (3 Lectures)
- **Supervised Learning: Fundamentals:** Learning from data, bias and variance, Model generalization- Overfitting and underfitting, Predictive modeling, data and target leakage, Data sets preparation for model building- Training, Testing and Validation datasets, cross validation, linear and logistic regression, K-nearest neighbor method, Business applications (appropriate case studies, use cases and situation analysis). (5 Lectures)
- **Supervised Learning: Decision Trees:** Fundamentals of decision trees, evaluating splits- information gain, Gini Index, applications (eg. Business use cases like Churn analysis/credit scoring etc.), Diversity and prediction; Ensemble of Learners- Multiple Models: Condorcet Jury Theorem and Diversity Prediction Theorem, random forest, AdaBoost, Model evaluation, comparison, interpretation and communication, Business applications (appropriate case studies, use cases and situation analysis). (8 Lectures)
- **Supervised Learning: Support Vector Machine (SVM):** Fundamentals of support vector machines (SVMs); Linear SVM classification- hard margin classification, soft margin classification; Non-linear separation- a step towards neural network, Predicting continuous variables, Business applications (appropriate case studies, use cases and situation analysis). (4 Lectures)
- **Unsupervised Learning:** Clustering Basics, Techniques used for clustering; k-means clustering- choosing k: elbow method, silhouette method; Density-based and hierarchical clustering methods; Principal Components analysis (PCA), Business applications (appropriate case studies, use cases and situation analysis). (5 Lectures)

Text Books

NA

References

1. Provost, F and Fawcett, T., **Data Science for Business**, Shroff Publishers and Distributors Pvt. Ltd, 2014.
2. Daniel S. Becker and Kai R. Larsen, **Automated Machine Learning for Business**, Oxford University Press, 2021.
3. Ethem Alpaydin, **Machine Learning**, MIT Press, 2016.
4. H. Brink, J.W. Richards and M. Fetheerolf, **Real-world Machine Learning**, Manning Pub., 2017.
5. J. Kelleher, B.M. Namee and A. D'Arcy, **Fundamentals of Machine Learning for Predictive Data Analytics**, MIT Press, 2015. (JK)
6. Scot Page, **The Model Thinker**, Basic Books, 2018.
7. K. Hosanagar, **A human guide to Machine intelligence, Portfolio**, Penguin Random House, 2019.
8. Matthew Kirk, **Thoughtful Machine Learning with Python**, Shroff Publishers and Distributors Pvt. Ltd, 2019.
9. Chip Huyen, **Designing Machine learning Systems**, Shroff Publishers and Distributors Pvt. Ltd., 2022.